Kaneda Fabrication Works (fictional)

WELDING PROCEDURE SPECIFICATION (WPS)

Format per ASME BPVC Section IX, Form QW-482 (see QW-200.1)

WPS no.	**WPS-07**
Revision	**1**
Date	**2025-02-14**
Supporting PQR	**PQR-07**

Welding process(es)	Type(s)
GTAW (root) + SMAW (fill and cap)	**Manual / Manual**

JOINTS (QW-402)

Joint design	Backing
Single-V groove, 60° included, root gap 2-3 mm, root face 1-2 mm	**None (open root, GTAW)**

BASE METALS (QW-403)

P-No.	Group No.	to P-No.	Group No.
1	**1 or 2**	**1**	**1 or 2**

Thickness range, base metal - groove	Fillet
5 to 50 mm	**All**

FILLER METALS (QW-404)

Process	Spec. No. (SFA)	AWS No. (Class)	F-No.	A-No.	Size	Deposited thickness
GTAW	SFA-5.18	ER70S-2	6	1	2.4 mm	max. 6 mm
SMAW	SFA-5.1	E7018	4	1	3.2 / 4.0 mm	max. 44 mm

POSITIONS (QW-405)

Position(s) of groove	Position(s) of fillet	Welding progression
All (1G-6G)	**All**	**Uphill**

PREHEAT (QW-406)

Preheat temperature, minimum	Interpass temperature, maximum
95 °C minimum	**250 °C maximum**

POSTWELD HEAT TREATMENT (QW-407)

PWHT	Temperature range	Time range
Required	**620 ± 15 °C (605 - 635 °C)**	**1 h per 25 mm of thickness, 15 min minimum**

GAS (QW-408)

GTAW: argon 99.99 %, 8-12 L/min. Backing gas: none.

ELECTRICAL CHARACTERISTICS (QW-409)

Weld pass(es)	Process	Classification	Diameter (mm)	Current / polarity	Amps (range)	Volts (range)	Travel (mm/min)
Root	GTAW	ER70S-2	2.4	DCEN	90 - 130	10 - 14	60 - 100
Fill	SMAW	E7018	3.2	DCEP	100 - 140	21 - 25	100 - 160
Cap	SMAW	E7018	4.0	DCEP	130 - 170	22 - 26	120 - 180

TECHNIQUE (QW-410)

String or weave bead: stringer, weave max. 3 x electrode diameter. Multiple passes per side.

Initial and interpass cleaning: brushing and grinding. Back gouging: grinding to sound metal.

Prepared by (welding engineer)	Approved by (QA manager)
K. Mori (fictional)	**L. Ortega (fictional)**
signed K. Mori 2025-02-14	*signed L. Ortega* 2025-02-14

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Kaneda Fabrication Works (fictional)

PROCEDURE QUALIFICATION RECORD (PQR)

Format per ASME BPVC Section IX, Form QW-483 · actual variables used to weld the test coupon

PQR no.	**PQR-07**
WPS qualified	**WPS-07**
Date welded	**2025-01-28**

TEST COUPON - ACTUAL VARIABLES	
Base metals (QW-403) **SA-516 Gr.70 to SA-516 Gr.70, 25 mm plate**	P-No. to P-No. **1 to 1**
Process(es) **GTAW (root) + SMAW (fill and cap)**	PWHT (QW-407) **620 °C, 60 min**

TENSILE TEST (QW-150)

Specimen	Width x thickness (mm)	Ultimate tensile (MPa)	Type of failure and location
T-1	19.0 x 25.0	535	Base metal, ductile
T-2	19.0 x 25.0	528	Base metal, ductile

GUIDED-BEND TESTS (QW-160)

4 side bends (QW-462.2), 10 mm, 180°: no open discontinuities. Acceptable.

TOUGHNESS TESTS (QW-170)

Specimen	Test temperature (°C)	Impact values
CVN 10 x 10	-29	Weld 54 / 60 / 57 J · HAZ 72 / 80 / 76 J

HARDNESS SURVEY (HV10)

Location	Maximum HV10
Weld metal and HAZ, cap and root	208

We certify that the statements in this record are correct and that the test welds were prepared, welded and tested in accordance with the requirements of Section IX of the ASME BPVC.

Mechanical tests by	Certified by (QA manager)
Halvorsen Test Lab (fictional) *signed Halvorsen Test Lab* Lab report HTL-07	**L. Ortega (fictional)** *signed L. Ortega* 2025-02-14

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